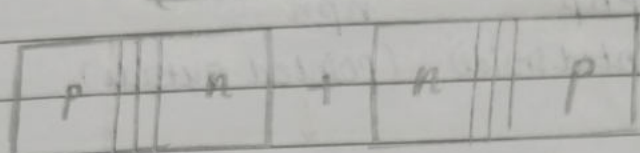
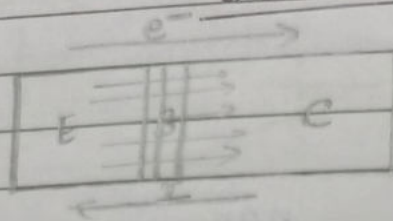
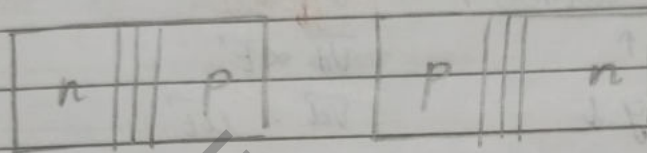


Transistor (current controlled device)



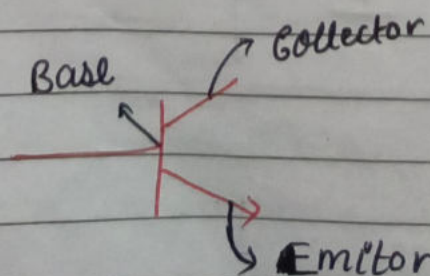
p-n-p transistor



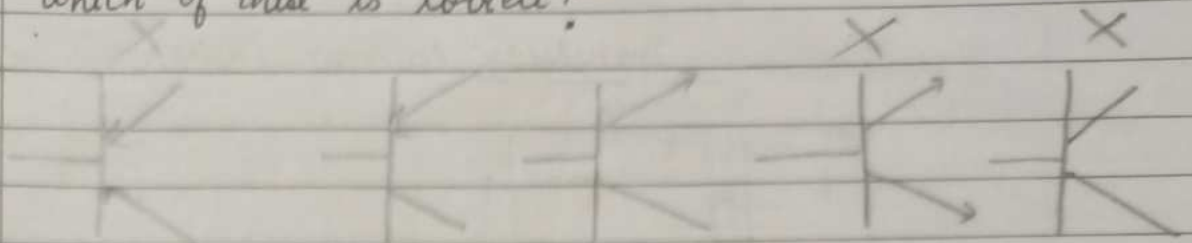
n-p-n transistor

Doping : $E > C > B$

Area : $C > E > B$



Ques Which of these is correct?



PNP
(pointed inside)

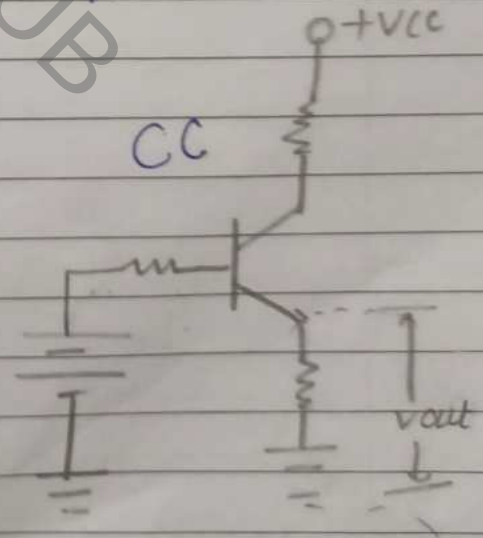
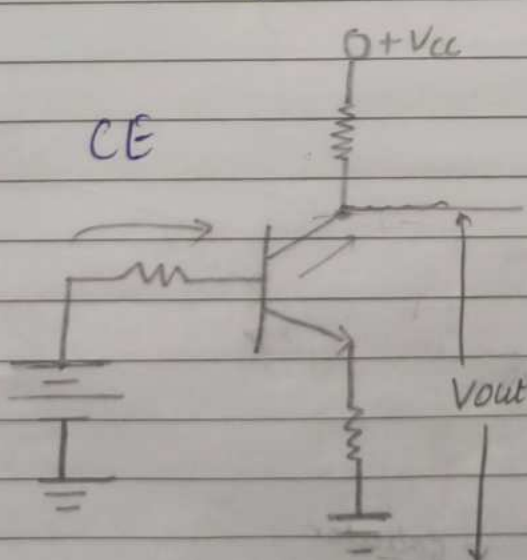
NPN
(pointed outside)

no. of charge carriers ↑
collision ↑
mobility ↓

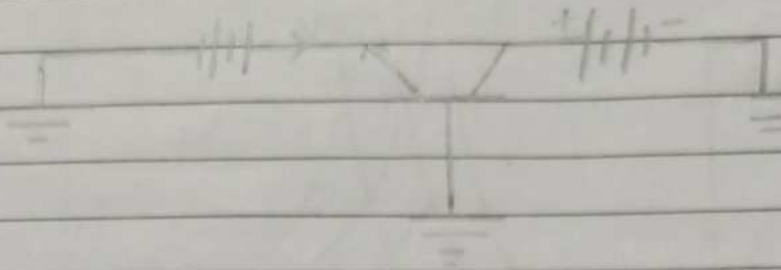
~~preferred~~
 $\bar{v}_d \propto E$
 $\bar{v}_d = \mu E$
 $\mu_n > \mu_p$
(Mobility)

Configuration of transistor

	I/P	O/P	
CB	E	C	$\alpha = \frac{I_C}{I_E}$
CE	B	C	$\beta = I_C / I_B$
CC	B	E	$\gamma = I_E / I_B$



Common Base



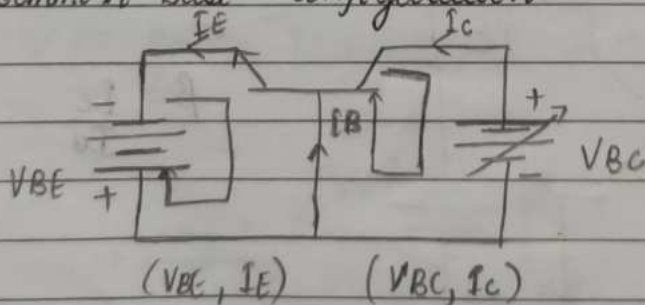
Mode of operation of BJT's

E	TE	B	IC	C
n	p	p	n	

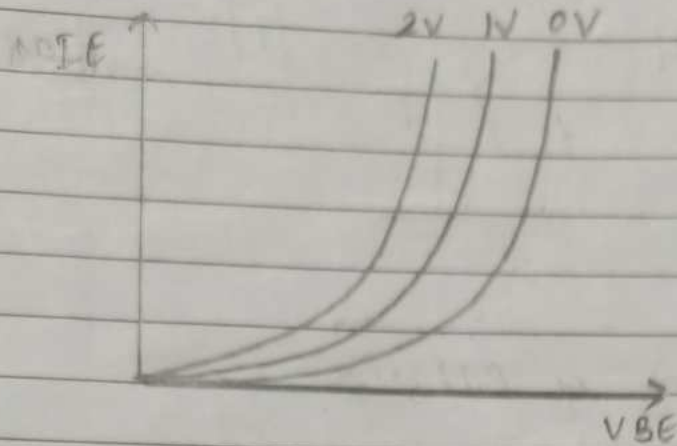
TE	IC	Mode of operation
Forward	Forward	Saturation
Forward	Reverse	Active
Reverse	Forward	Reverse Active
Reverse	Reverse	Cut off region

★ Configuration of BJT's

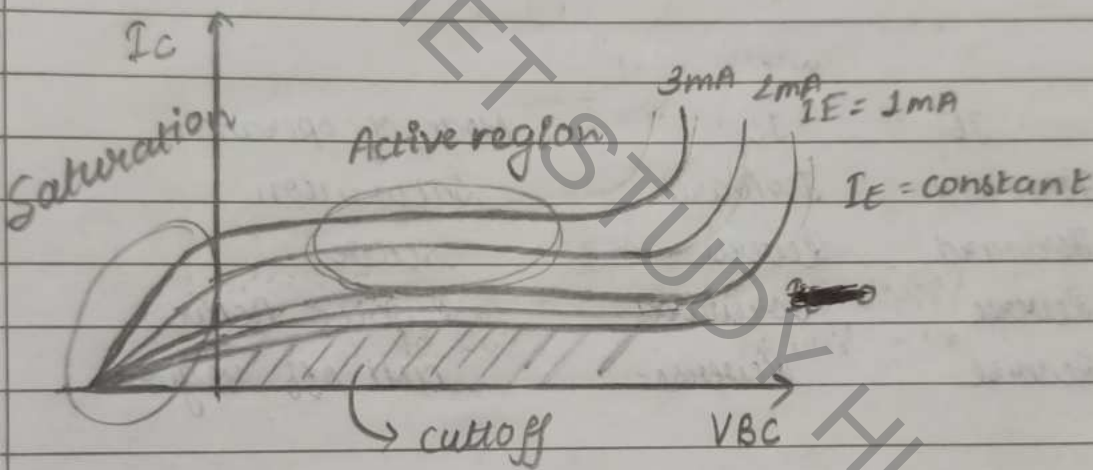
Common Base Configuration



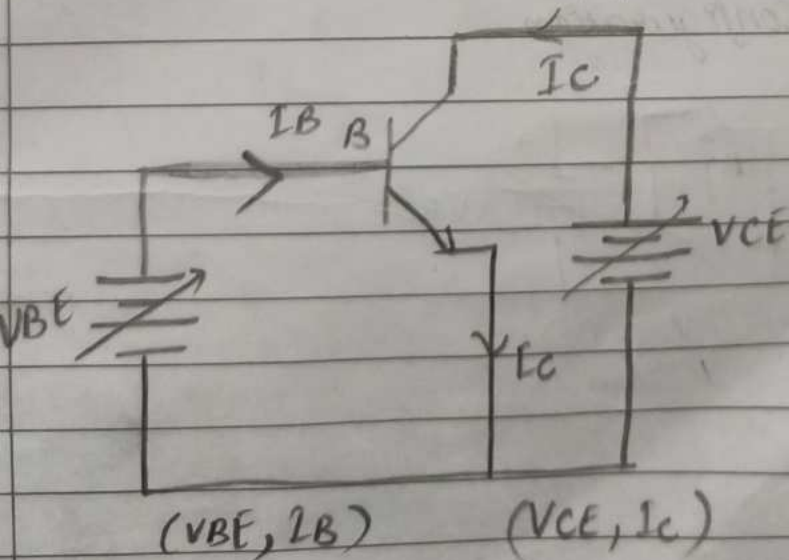
Input Characteristics



Output characteristics

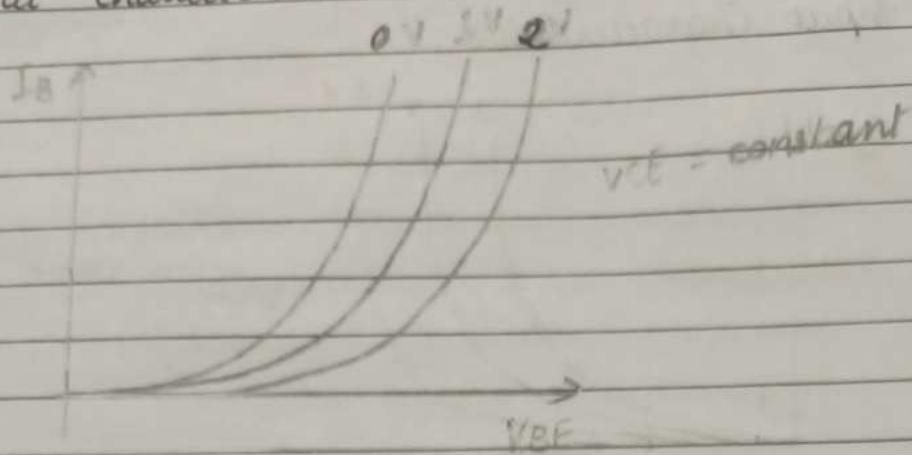


Common emitter configuration

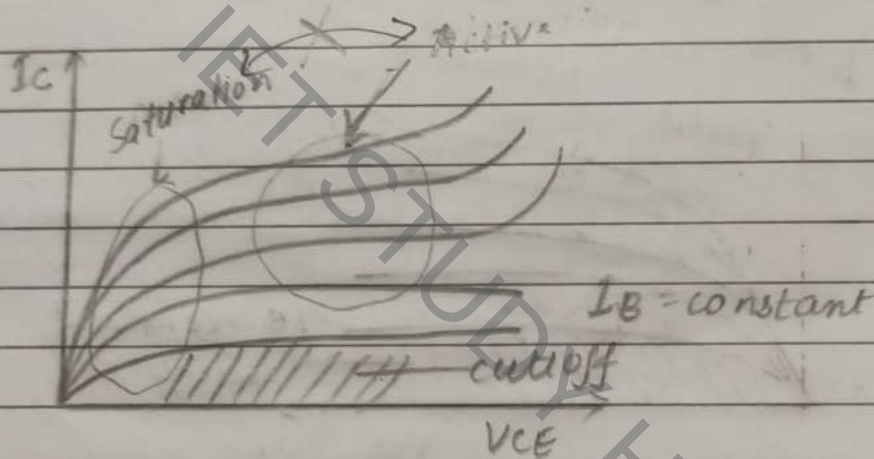


$$\beta = \frac{I_C}{I_B}$$

Input characteristics

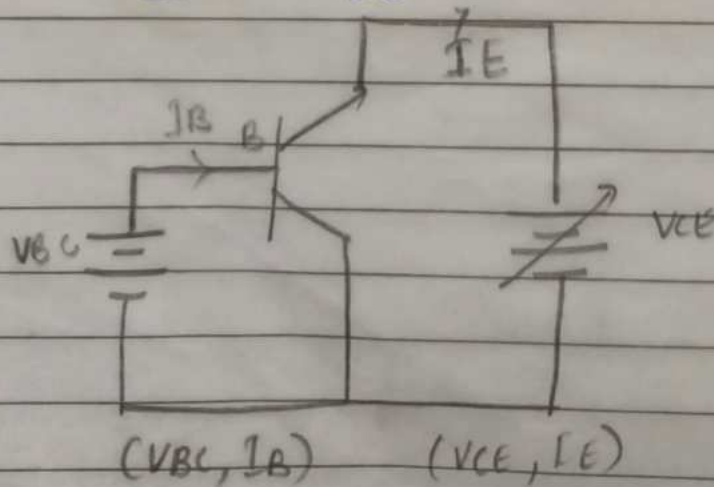


Output characteristics

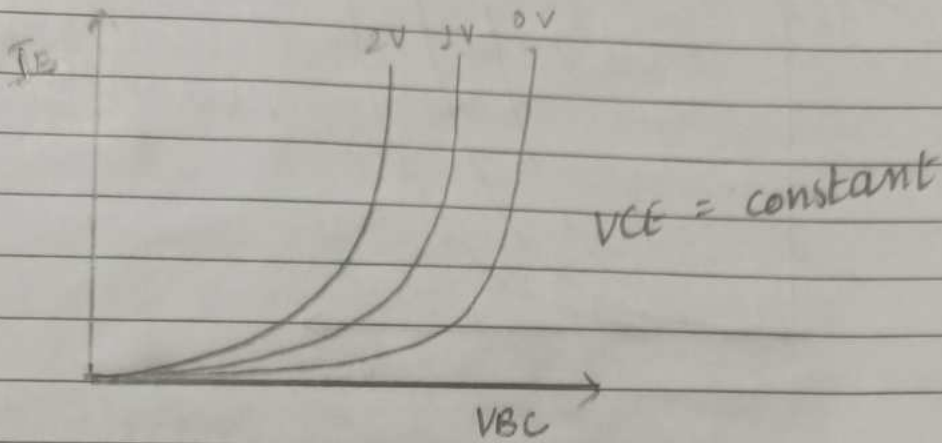


Common collector configuration

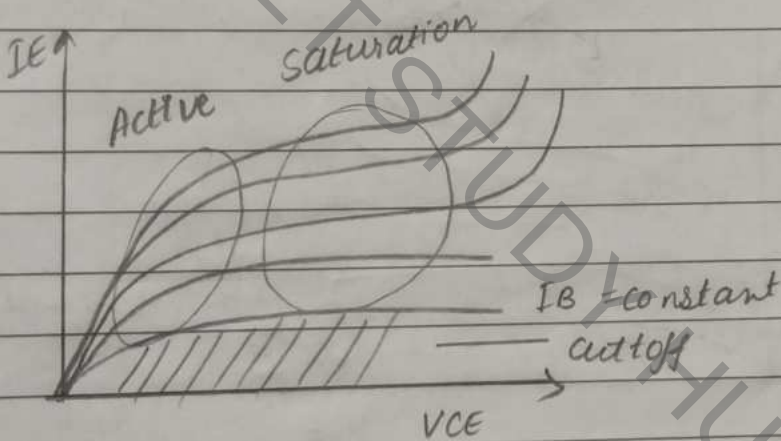
$$\beta = \frac{I_E}{I_B} = \frac{I_C + I_B}{I_B} = \beta + 1$$



Input Characteristics



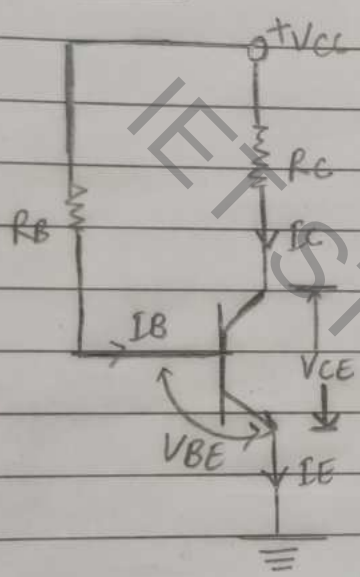
Output Characteristic



Biasing of a transistor (CE)

- ★ Fixed biased
- ★ Fixed biased with emitter resistor
- ★ self biased
- ★ self biased with emitter resistor
- ★ voltage divider biased

1. Fixed Biased



Input side,

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B R_B = V_{CC} - V_{BE}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$\frac{I_C}{I_B} = \beta$$

$$I_C = \beta I_B = \beta \left(\frac{V_{CC} - V_{BE}}{R_B} \right)$$

Output side,

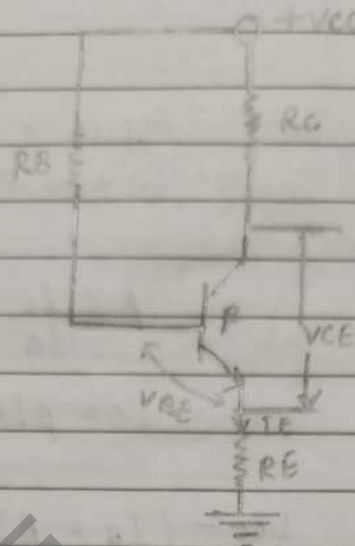
$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_C$$

$$= V_{CC} - \beta \left(\frac{V_{CC} - V_{BE}}{R_B} \right) R_C$$

Temp ↑ β ↑ I_C ↑ Temp ↑

2. Fixed Bias with emitter resistor



Input

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$- I_E R_E = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$V_{CC} - I_B R_B - V_{BE} - (\beta + 1) I_B R_E = 0$$

$$I_B R_B + (\beta + 1) I_B R_E = V_{CC} - V_{BE}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E}$$

$$I_C = \beta I_B = \beta \left(\frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E} \right)$$

$$= \left(\frac{\beta}{\beta + 1} \right) \left(\frac{V_{CC} - V_{BE}}{\frac{R_B}{\beta + 1} + R_E} \right)$$

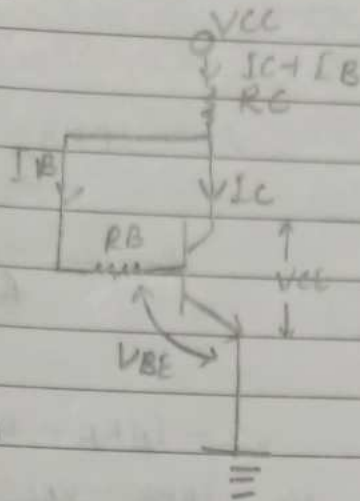
$$\beta \cong \beta + 1, \quad R_E \gg \frac{R_B}{\beta + 1}$$

$$I_C \cong \frac{V_{CC} - V_{BE}}{R_E}$$

Output

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

3. self biased



Input $\rightarrow$

$$V_{CC} - (I_C + I_B)R_C - I_B R_B - V_{BE} = 0$$

$$\beta = \frac{I_C}{I_B}$$

$$I_C = \beta I_B$$

$$V_{CC} - (I_B + \beta I_B)R_C - I_B R_B - V_{BE} = 0$$

$$V_{CC} - V_{BE} = (1 + \beta)I_B R_C + I_B R_B$$

$$V_{CC} - V_{BE} = I_B (R_C + \beta R_C + R_B)$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_C + \beta R_C + R_B}$$

$$I_C = \beta I_B = \beta \left(\frac{V_{CC} - V_{BE}}{R_C + \beta R_C + R_B} \right)$$

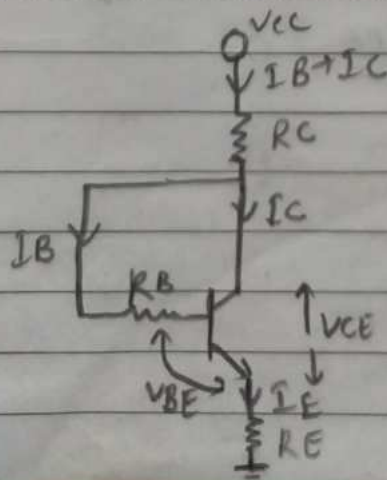
Output $\rightarrow$

$$V_{CC} - (I_B + I_C)R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_C - I_B R_C$$

$$V_{CE} = V_{CC} - \beta \left(\frac{V_{CC} - V_{BE}}{R_C + \beta R_C + R_B} \right) R_C - \left(\frac{V_{CC} - V_{BE}}{R_C + \beta R_C + R_B} \right) R_C$$

4. self Biased with emitter



Input $\rightarrow$

$$V_{CC} - (I_B + I_C)R_C - I_B R_B - V_{BE} - I_E R_E = 0$$

$$V_{CC} - (I_B + \beta I_B)R_C - I_B R_B - V_{BE} - (\beta + 1)I_B R_E = 0$$

$$V_{CC} - V_{BE} = I_B (R_C + \beta R_C + R_B + \beta R_E + R_E)$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_C + \beta R_C + R_B + \beta R_E + R_E}$$

$$I_C = \beta \left(\frac{V_{CC} - V_{BE}}{R_C + \beta R_C + R_B + \beta R_E + R_E} \right)$$

$$= \frac{\beta}{(\beta+1)} \left(\frac{V_{CC} - V_{BE}}{R_C + R_E + \frac{R_B}{(\beta+1)}} \right) \approx \frac{V_{CC} - V_{BE}}{R_C + R_E}$$

Output $\rightarrow$

$$V_{CC} - (I_B + I_C) R_C - V_{CE} - I_E R_E = 0$$

$$V_{CE} = V_{CC} - I_B R_C - I_C R_C - I_E R_E$$

$$V_{CE} = V_{CC} - \left(\frac{V_{CC} - V_{BE}}{R_C + \beta R_C + R_B + \beta R_E + R_E} \right) R_C$$

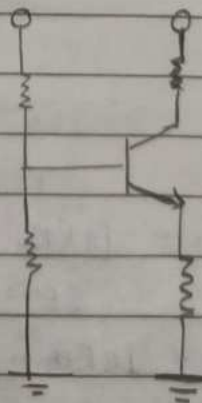
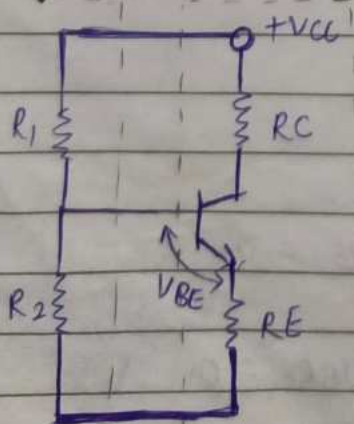
$$- \beta \left(\frac{V_{CC} - V_{BE}}{R_C + \beta R_C + R_B + \beta R_E + R_E} \right) R_C$$

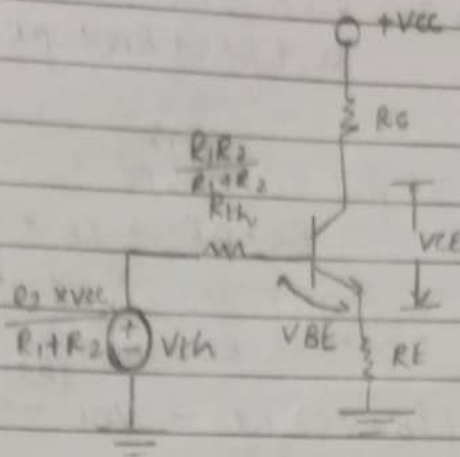
$$- (\beta + 1) \left(\frac{V_{CC} - V_{BE}}{R_C + \beta R_C + R_B + \beta R_E + R_E} \right) R_E$$

$$V_{CE} = V_{CC} - \left(\frac{V_{CC} - V_{BE}}{R_C + \beta R_C + R_B + \beta R_E + R_E} \right) R_C$$

$$- \left(\frac{V_{CC} - V_{BE}}{R_C + R_E} \right) R_C - \left(\frac{V_{CC} - V_{BE}}{R_C + R_E} \right) R_E$$

5. Voltage divider biased





$$\begin{aligned}
 V_{th} - I_B R_{th} - V_{BE} - I_E R_E &= 0 \\
 I_E &= (\beta + 1) I_B \\
 V_{th} - I_B R_{th} - V_{BE} - (\beta + 1) I_B R_E &= 0 \\
 I_B &= \frac{V_{th} - V_{BE}}{R_{th} + (\beta + 1) R_E}
 \end{aligned}$$

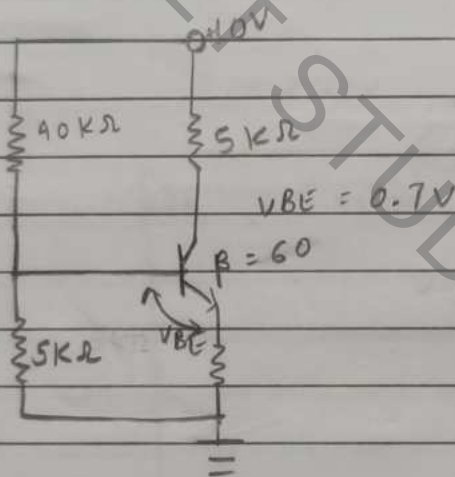
$$I_C = \beta I_B = \beta \left(\frac{V_{TH} - V_{BE}}{R_{TH} + (\beta + 1)R_E} \right)$$

$$I_C = \left(\frac{\beta}{\beta + 1} \right) \left(\frac{V_{TH} - V_{BE}}{\frac{R_{TH}}{\beta + 1} + R_E} \right)$$

$$R_E \gg \frac{R_{TH}}{\beta + 1}$$

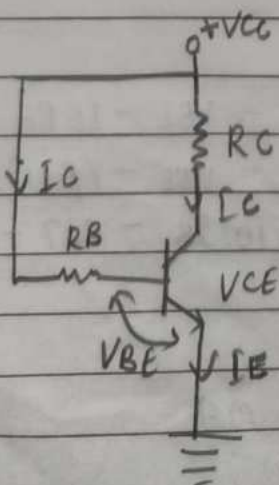
$$I_C \cong \frac{V_{TH} - V_{BE}}{R_E}$$

Ques



Ans

Determine Q point of transistor, assume that $V_{CC} = 8V$, $R_B = 360k\Omega$, $R_C = 2k\Omega$, β (current gain) = 100. Also draw loadline.



$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$8 - I_B \times 360 \times 10^3 - 0.7 = 0$$

$$I_B = \frac{8 - 0.7}{360 \times 10^3} = 20.28 \mu A$$

$$\begin{aligned} I_C &= \beta I_B = 100 \times 20.28 \mu A \\ &= 20.28 \times 10^{-4} A \\ &= 2.028 mA \end{aligned}$$

Q (3.5 Date 4V 2.028mA)

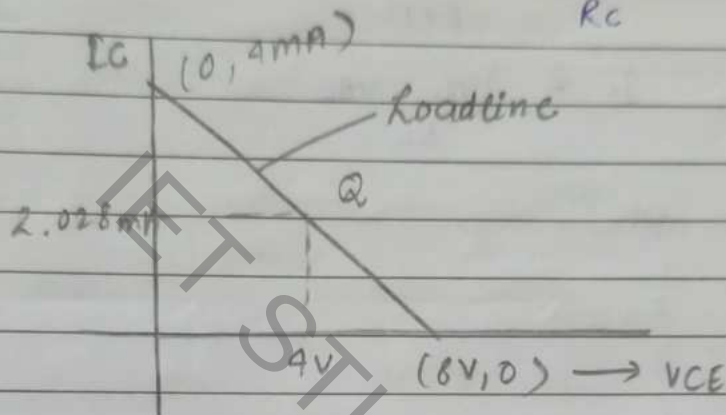
$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$8 - 2.028 \times 10^{-3} \times 2 \times 10^3 - V_{CE} = 0$$

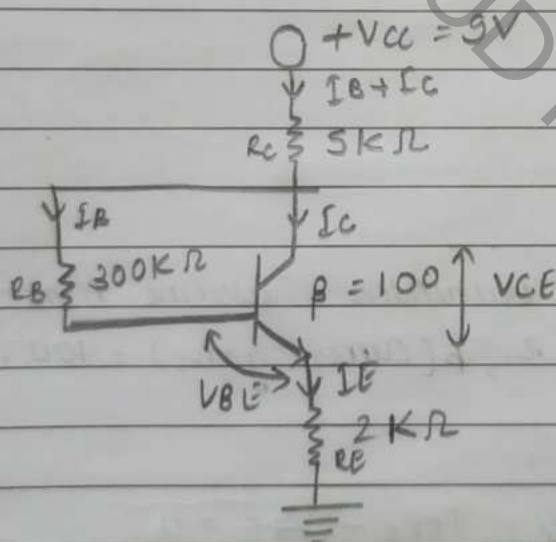
$$V_{CE} = 3.94V$$

When $I_C = 0$, $V_{CE} = V_{CC} = 8V$

When $V_{CE} = 0$, $I_C(\text{sat}) = \frac{V_{CC}}{R_C} = \frac{8V}{2 \times 10^3} = 4mA$



Ans



$$V_{CC} - (I_B + I_C) R_C - I_B R_B - V_{BE} - I_E R_E = 0$$

$$V_{CC} - (I_B + \beta I_B) R_C - I_B R_B - V_{BE} - (\beta + 1) I_B R_E = 0$$

$$9 - 101 I_B \times 5 \times 10^3 - 300 \times 10^3 I_B - 0.7 - (101) 2 \times 10^3 I_B = 0$$

$$9 - 0.7 = I_B$$

$$1007 \times 10^3$$

$$I_B = 0.00824 mA$$

$$I_B = 8.24 \mu A$$

$$I_C = \beta I_B = 100 \times 8.24 \mu A = 0.824 \text{ mA}$$

Output side

$$V_{CC} - I_E \times 5 \times 10^3 - V_{CE} - I_E \times 2 \times 10^3 = 0$$

$$9 - 101 \times 8.24 \times 10^{-6} \times 5 \times 10^3 - V_{CE} - 101 \times 8.24 \times 10^{-6} \times 2 \times 10^3 = 0$$

$$\begin{aligned} V_{CE} &= 9 - 5825.68 \times 10^{-3} \\ &= 9 - 5.82 \\ &= 3.18 \text{ V} \end{aligned}$$

$$Q(3.18 \text{ V}, 0.824 \text{ mA})$$

when $I_C = 0$,